

Feature 1: Audit Log Date Range Filter Validation [FR-4/ Auditability]

Requirement: In the Audit Log viewer, the administrator can filter file distribution and download history by date range (Start Date to End Date). The system requires that Start Date $\leq$ End Date and neither date can be in the future.

Detailed Test Case Specification

Field	Details
Test Case ID	TC-DLD-EP-08
Test Scenario	Verify that the audit log query module rejects queries where the Start Date is later than the End Date (FR-4 / EP).
Preconditions	1. Administrator is logged in and viewing the Audit Trail Dashboard (/admin/audit-logs).
Test Steps	1. Select Start Date: 2026-10-08. 2. Select End Date: 2026-09-01. 3. Click "Apply Filter".
Input Data	- Start Date: 2026-10-08 - End Date: 2026-09-01
Expected Result	System blocks the database query and displays error message: "Invalid date filter: Start date cannot be later than end date." Log grid remains unchanged.
Actual Result	(To be completed during test execution)
Status	(Pass / Fail)
Remarks	Validates inverted date partition P3 using Equivalence Partitioning (EP).

Feature 2: File Access / Download Expiration Duration [FR-2/NFR-1] (BVA)

Requirement: When sharing a confidential file, the administrator sets an access validity period between 1 and 30 days. Values of 0 days or exceeding 30 days must be rejected.

Designed Test Case Table

Field	Details
Test Case ID	TC-DLD-BVA-07
Test Scenario	Verify that the system blocks configuring file access expiration exceeding 30 days during file distribution (FR-2/NFR-1).
Preconditions	1. Administrator is on the File Distribution settings screen. 2. File quarterly_financials.txt is selected for sharing.
Test Steps	1. In the "Access Expiration (Days)" field, enter 31. 2. Select target agents. 3. Click "Publish Distribution".
Input Data	- Access Expiration Days: 31
Expected Result	System displays field validation error: "Invalid validity period: File access duration must be between 1 and 30 days." Form submission is blocked.
Actual Result	(To be completed during test execution)
Status	(Pass/Fail)
Remarks	Validates upper boundary value $30+1=31$ days using Boundary Value Analysis (BVA).

Feature 3: Batch Recipient Email Notification Limit [FR-2/ NFR-3]

Requirement: When a new confidential file is distributed, the backend triggers automatic email notifications to allocated agents. To prevent SMTP server throttling and denial of service, the notification dispatcher processes a maximum of 10 recipient email addresses per outgoing SMTP transaction batch.

Detailed Test Case Specification

Field	Details
Test Case ID	TC-DLD-BVA-09
Test Scenario	Verify that email notification batch dispatch correctly enforces the 10-recipient boundary threshold and queues surplus recipients into a secondary batch (FR-2 / NFR-3/ BVA).
Preconditions	1. Administrator initiates distribution of quarterly_audit.txt to 11 authorized agents. 2. SMTP Notification Service is active.
Test Steps	1. Confirm allocation of 11 agents. 2. Submit distribution workflow. 3. Inspect backend notification dispatcher logs.
Input Data	- Total Recipients: 11 agents (agent1@org.com to agent11@org.com)
Expected Result	Dispatcher packages first 10 recipients into Batch 1, dispatches immediately, and queues the 11th recipient into Batch 2, preventing SMTP rate limit rejection.
Actual Result	(To be completed during test execution)
Status	(Pass / Fail)
Remarks	Validates upper boundary + 1 threshold (10+1=11) using Boundary Value Analysis (BVA).